

Abstract (English)

Hypertension is a top risk factor causing heart-related problems, especially in older people, and thus, constant BP monitoring is necessary. Cuff-less methods of BP estimation based on physiological signals and artificial intelligence have been developed because of inconvenience caused by traditional BP measuring devices to constantly measure BP.

This thesis introduces SENTINEL (Smart Elderly Navigation & Telehealth Integrated Lifeline), an intelligent health monitoring platform which utilizes deep learning-based cuff-less blood pressure estimation and a web-based monitoring system designed specifically for elderly patients and their families. In this framework, systolic and diastolic blood pressures will be estimated in parallel by using multi-task learning techniques.

In this thesis, we examine three physiological signals (ECG, PPG and RESP) in combination with various deep learning architectures (CNN, RNN, LSTM, GRU, Bi-LSTM) to find optimal combinations of modality and architecture. Based on these results, multimodal learning will be tested for dual-modalities and tri-modalities in conjunction with four different fusion techniques: feature concatenation, bidirectional cross-attention, Gated Multimodal Unit (GMU), and a proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF).

The experimental validation on the public MIMIC-BP dataset showed that PPG is the best modality in terms of single-modality prediction. It was proved that the proposed DGAGF framework efficiently combines complementary information coming from other modalities. In addition, the combination of PPG and RESP turned out to be the best multimodal combination and outperformed a conventional combination of ECG and PPG. It was shown that proper fusion methods have a stronger influence on prediction results than the increase in the number of used modalities.

In order to make the proposed framework more applicable, the trained AI models were incorporated in the designed monitoring application implemented with the help of React, FastAPI, and SQLite, allowing for patient management, prediction history, alerts generation, and remote monitoring of relatives.